

Electromagnetic Theory (SC217)

December 4, 2025

1 Course Placement

Electromagnetic Theory (SC217) is a compulsory course for BTech (ICT and ICT-CS) students.

2 Course Format

The course will consist of 3 hours of lectures each week, and one tutorial of duration 1 hour.

3 Grading Policy

Grades will be based on two mid-sem/in-sem examinations and the end-semester examination. The weightage will be 25%, 25% and 50% respectively.

4 Course Outcomes

This course is targeted at engineering students and will teach basic electrodynamics which governs the response of a medium to a signal. The course will mainly concentrate on electromagnetic waves and its behaviour in different media. At the conclusion of the course, students should be able to,

- To understand and analyse different electromagnetic phenomena.
- To understand the principles of communication devices which utilise electromagnetic waves.
- Understand and analyze complicated engineering and real-world situations by physical reasoning in terms of simple fundamental physical laws.
- Formulate mathematical models by applying abstract concepts to complex problems and solving the model using approximations if necessary.

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
x	x	x	x								x

5 Syllabus and lecture sequence

1. Electromagnetism and electrodynamics. Statics and dynamics of charged particles. Nature of charges.
2. Scales of interaction between charges. Observations about electric and magnetic fields by Coulomb, Oersted, Ampere and Faraday. Unification of electricity, magnetism and optics by Maxwell.
3. Forces of interaction, direct action and action at a distance. Concept of a field. Scalar and vector fields.
4. Flux, flux density and field strength. Local and global descriptions of field variations. Derivatives of scalar functions of one and higher-dimensional systems.

5. Vector dot product, the line element, the “del” operator, ∇ . The gradient operation on a scalar. Physical interpretation of gradients and the extremum condition.
6. Divergence and curl of vector fields. Physical interpretation of the divergence and the curl operators.
7. Identities of second derivatives – the divergence of a curl and the curl of a gradient.
8. Potentials and conservative systems.
9. Line element, surface element and volume element. Vector integrals.
10. Theorem of gradients. The Gauss divergence theorem and the Stokes curl theorem. Implications for solenoidal, irrotational and conservative fields.
11. Curvilinear coordinates: The cylindrical coordinate system. The unit vectors. The line element, the surface elements and the volume element. Orthogonality of the unit vectors.
12. Curvilinear coordinates: The spherical polar coordinate system. The unit vectors. The line element, the surface elements and the volume element. Orthogonality of the unit vectors.
13. Vector differential operators (the gradient, the divergence and the Laplacian) in the curvilinear systems.
14. Electrostatics, Coulomb’s law and the principle of superposition. Flux of isolated and multiple charges.
15. Integral and differential forms of Gauss’s law of electrostatics.
16. Gauss’s law in spherical symmetry — hollow, solid and concentric hollow spheres.
17. Gauss’s law in cylindrical symmetry — long conductor, hollow, solid and concentric hollow cylinders.
18. Gauss’s law in plane symmetry. Electrostatic boundary conditions. Properties of conductors. Surface charge and the force on a conductor.
19. Poisson’s and Laplace’s equations. Potential formalism, points of reference and potentials due to isolated and multiple charges. Superposition of potential. Work.
20. Poisson’s equation in planar, cylindrical and spherical symmetries.
21. Laplace’s equation, its properties and boundary conditions. The first and second uniqueness theorems.
22. Laplace’s equation in planar symmetry, cylindrical symmetry (with s , ϕ and z -dependence) and spherical symmetry (with r , θ and ϕ -dependence).
23. The method of images, induced surface charge, force and energy.
24. Dipole moment and the potential of a dipole. Dielectrics, polarisability, polarisation, bound charges.
25. Electric displacement, linear dielectrics, susceptibility and relative permittivity.
26. Magnetostatics, the Lorentz force, work by magnetic forces, currents, the Biot-Savart law and its application on a long straight current.
27. Integral and differential forms of Ampere’s law.
28. Applications of Ampere’s law — long conducting wire, long solid conductor, long hollow sheath and long concentric sheaths.
29. Vanishing of magnetic flux and divergence, magnetostatic boundary conditions, Maxwell’s equations for electrostatics and magnetostatics.
30. Electromotive force, changing magnetic fields, magnetic flux, Faraday’s law.

31. Current flux, the equation of continuity, Maxwell's correction of Ampere's law, displacement current, Maxwell's equations of electrodynamics,
32. Magnetic fields due to a current-carrying loop, current loop formed by an electron
33. Polarisation current, Maxwell's equations in matter and in free space.
34. The wave equation in free space from Maxwell's equations and light as an electromagnetic wave. Electromagnetic waves in matter, refractive index.
35. Plane electromagnetic waves, transverse oscillations of orthogonal electric and magnetic fields.
36. The electromagnetic energy density, the continuity equation and the Poynting vector. Average energy density and irradiance.
37. Momentum density and radiation pressure on absorbing and reflecting surfaces. Electromagnetic boundary conditions at interfaces.
38. Reflection and transmission at normal incidence. Reflection and transmission coefficients.
39. Oblique incidence. The basic laws of geometrical optics. Total internal reflection.

6 Textbooks and References

1. *Introduction to Electrodynamics*, David J. Griffiths, Pearson Prentice Hall, Third Edition.
2. *Principles Of Electromagnetics*, Mathew N. O. Sadiku and S. V. Kulkarni, McGraw Hill, 7th Edition.
3. *Electromagnetic Waves and Radiating Systems*, Edward C. Jordan and Keith G. Balmain, Pearson, Second Edition.

Other reference books/material will be suggested in class from time to time, as appropriate.

7 Course Credits

(3 – 1 – 0 – 4)

8 Evaluation

Evaluation will be on the basis of performance in two in-semester and one end-semester examination.